Query Device Information

Objective: Learn how to query and display device information using the get_info template.

Task:

1. List all available devices and display their names, global memory sizes, and the number of compute units.
2. Print the information to the console.

```
#include <sycl/sycl.hpp>
#include <iostream>

using namespace sycl;

int main() {
    for (const auto &platform : platform::get_platforms()) {
        for (const auto &dev : platform.get_devices()) {
            std::cout << "Device: " << dev.get_info<info::device::name>() << std::endl;
            std::cout << "  Global Memory Size: " << dev.get_info<info::device::global_mem_size>() << " bytes" << std::endl;
            std::cout << "  Max Compute Units: " << dev.get_info<info::device::max_compute_units>() << std::endl;
        }
    }
    return 0;
}
```

Create and Use Custom Device Selector

Objective: Learn how to create a custom device selector to select a specific type of device.

Task:

1. Create a custom device selector that prefers Intel GPUs.
2. Use this selector to create a queue and print the selected device's name.

```
#include <sycl/sycl.hpp>
#include <iostream>

using namespace sycl;

class special_device_selector : public device_selector {
public:
    int operator()(const device &dev) const override {
        if (dev.is_gpu()) {
```

```
            auto vendorName = dev.get_info<info::device::vendor>();
            if (vendorName.find("Intel") != std::string::npos) {
                return 1;
            }
        }
        return -1;
    }
};

int main() {
    special_device_selector selector;
    queue Q(selector);

    auto dev = Q.get_device();
    std::cout << "Selected Device: " << dev.get_info<info::device::name
        >() << std::endl;
    return 0;
}
```

Using Aspects for Device Selection

Objective: Learn how to use aspects to select devices with specific capabilities.

Task:

1. Use an aspect selector to select devices that support unified shared memory (USM) and FP16 capabilities.
2. Create a queue with the selected device and print the device's name.

```
#include <sycl/sycl.hpp>
#include <iostream>

using namespace sycl;

int main() {
    auto my_selector = aspect_selector(aspect::usm_device_allocations,
        aspect::fp16);
    queue Q(my_selector);

    auto dev = Q.get_device();
    std::cout << "Selected Device: " << dev.get_info<info::device::name
        >() << std::endl;
    return 0;
}
```